

Notes on lecture 16

Jiten Chandra Kalita and Shyamashree Upadhyay

1 Identifying conservative vector fields

Question: How is it possible to determine whether or not a vector field $\vec{F}$ is conservative?

Suppose it is known that $\vec{F} = P\hat{i} + Q\hat{j}$ is conservative, where P and Q have continuous first order partial derivatives. Then there exists a function f such that $\vec{F} = \nabla f$, that is,

$$P = \frac{\partial f}{\partial x} \text{ and } Q = \frac{\partial f}{\partial y}.$$

Therefore, by Clairaut's theorem,

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}.$$

This proves the following theorem:

Theorem 1.0.1. *If $\vec{F}(x, y) = P(x, y)\hat{i} + Q(x, y)\hat{j}$ is a conservative vector field, where P and Q have continuous first order partial derivatives on a domain D , then throughout D we have*

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Remark 1.0.2. The converse of the above theorem (Theorem 1.0.1) is true only for a special type of region. But to explain that, we need the concept of a simply-connected region.

We had found out that if $\vec{F} = P\hat{i} + Q\hat{j}$ is a conservative vector field, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}. \quad (*)$$

A natural question arises now, namely, if $(*)$ holds, then is $\vec{F} = P\hat{i} + Q\hat{j}$ conservative?

To answer this question, we need to learn some concepts first.

Definition 1.0.3. If a curve (in plane or in space) does not cross itself, then it is called a **simple curve**. Suppose a curve is parametrized by $\vec{r}(t)$, where $a \leq t \leq b$. If the curve is NOT simple, then there exists $t_1, t_2 \in (a, b)$ such that

$$a < t_1 < t_2 < b \text{ and } \vec{r}(t_1) = \vec{r}(t_2).$$

□

Definition 1.0.4. A **simple closed curve** is a curve that is simple as well as closed. □

Figure 1.0.1 illustrates the above two definitions.

Definition 1.0.5. Consider a region D in $\mathbb{R}^2$. If any simple closed curve C in D encloses points belonging to D only, then we say that the region D is **simply connected**. □

Figure 1.0.2 illustrates the above definition of a simply connected region.

Remark 1.0.6. Observe that in the 3 pictures towards the bottom of Figure 1.0.2, we have regions that are not simply connected. This is because there exists curves in those regions which enclose points which are not in the region. Intuitively, a region is NOT simply connected if it has some “holes” in it.

Now comes the theorem which answers the question (asked at the beginning of this lecture) about conservative vector fields.

Theorem 1.0.7. Let $\vec{F} = P\hat{i} + Q\hat{j}$ be a vector field on an open simply connected region D . Suppose that P and Q have continuous first order partial derivatives and

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

throughout D . Then $\vec{F}$ is conservative.

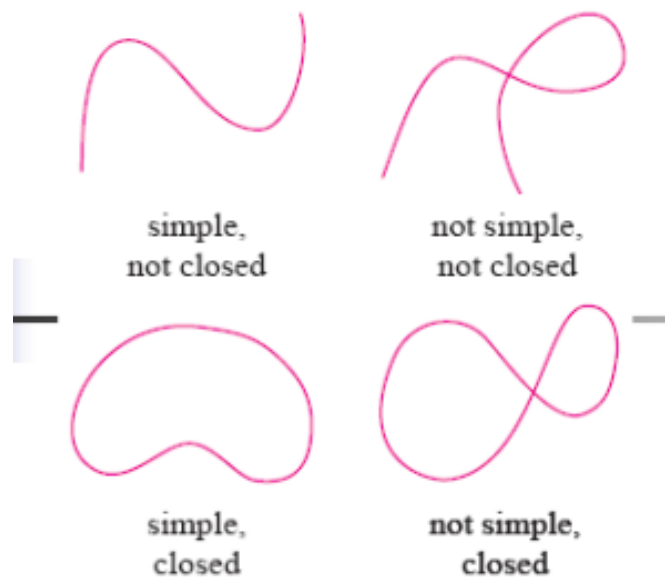


Figure 1.0.1: Simple and NOT simple curves

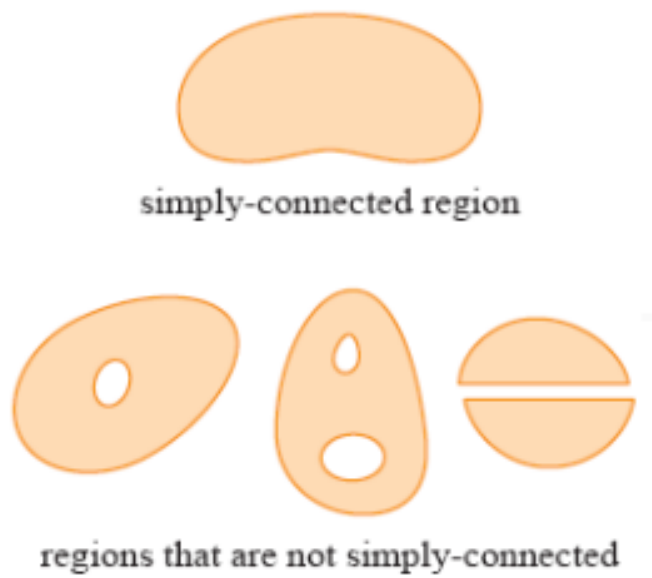


Figure 1.0.2: Simply connected region

PROOF: Omitted. □

Remark 1.0.8. In theorem 1.0.9 above, we are considering vector fields on $\mathbb{R}^2$ only because here we are having the condition $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$. There is no z variable involved in this.

Example 1.0.9. Let $\vec{F} = (x - y)\hat{i} + (x - 2)\hat{j}$.
Observe that $\frac{\partial P}{\partial y} = -1$ and $\frac{\partial Q}{\partial x} = 1$. So

$$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}.$$

Hence $\vec{F}$ is not conservative. [Here we had considered the region D to be the entire $\mathbb{R}^2$.] □

Example 1.0.10. Let $\vec{F} = (3 + 2xy)\hat{i} + (x^2 - 3y^2)\hat{j}$.
Here it is easy to check that the condition

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

is satisfied. But we need to check whether or not the region on which the vector field $\vec{F}$ is being considered, is open and simply connected.

We can consider the vector field $\vec{F}$ over the entire $\mathbb{R}^2$, which is an open simply connected region. So $\vec{F}$ is a conservative vector field on $\mathbb{R}^2$. That means there exists a scalar function f such that

$$\vec{F} = \nabla f.$$

To find f , observe that

$$\vec{F} = \nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}.$$

Then we have

$$\frac{\partial f}{\partial x} = 3 + 2xy \tag{1}$$

and

$$\frac{\partial f}{\partial y} = x^2 - 3y^2. \tag{2}$$

If we integrate (1) partially with respect to x , then we will get

$$f(x, y) = 3x + x^2y + \phi(y),$$

for some function $\phi(y)$ of y . We get the $\phi(y)$ upon integration because y is treated as a constant in this partial integration. Then we have

$$\begin{aligned} f_y &= x^2 + \phi'(y) = x^2 - 3y^2 \\ \Rightarrow \phi'(y) &= -3y^2 \\ \Rightarrow \phi(y) &= -y^3 + C, \end{aligned}$$

for some constant C . Hence

$$f(x, y) = 3x + x^2y - y^3 + C.$$

Now suppose we want to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is given by

$$\vec{r}(t) = e^t \sin t \hat{i} + e^t \cos t \hat{j}, \quad 0 \leq t \leq \pi.$$

If we compute it directly, then we have

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_C Pdx + Qdy \\ &= \int_0^\pi (3 + 2e^{2t} \sin t \cos t)(e^t \sin t + e^t \cos t)dt + (e^{2t} \sin^2 t - 3e^{2t} \cos^2 t)(e^t \cos t - e^t \sin t)dt. \end{aligned}$$

But this is a lengthy computation.

On the other hand, since $\vec{F}$ is conservative, we can use the fundamental theorem of line integrals and get

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \nabla f \cdot d\vec{r} = f(\vec{r}(\pi)) - f(\vec{r}(0)).$$

Note that

$$\begin{aligned} f(x, y) &= 3x + x^2y - y^3 + C \\ &= 3e^t \sin t + e^{3t} \sin^2 t \cos t - e^{3t} \cos^3 t + C. \end{aligned}$$

Then

$$f(\vec{r}(\pi)) - f(\vec{r}(0)) = [e^{3\pi} + C] - [-1 + C] = e^{3\pi} + 1.$$

This is how we could compute the same integral in a much more easier way. This is the utility of $\vec{F}$ being conservative and the fundamental theorem of line integrals. $\square$

2 Green's theorem

We need to learn some concepts first for studying Green's theorem.

Definition 2.0.1. Consider a simple closed curve C which bounds a region D . Whenever you move along the curve C (clockwise/anti-clockwise), if the region D is towards your left, then we call C to be a **positively oriented curve**. $\square$

Remark 2.0.2. Suppose we consider a region which looks like an annulus, bounded by 2 curves, one from outside and another from inside. Then the inner curve should be oriented in the clockwise direction to obtain the region D on your left.

Figure 2.0.1 shows some positively and negatively oriented curves.

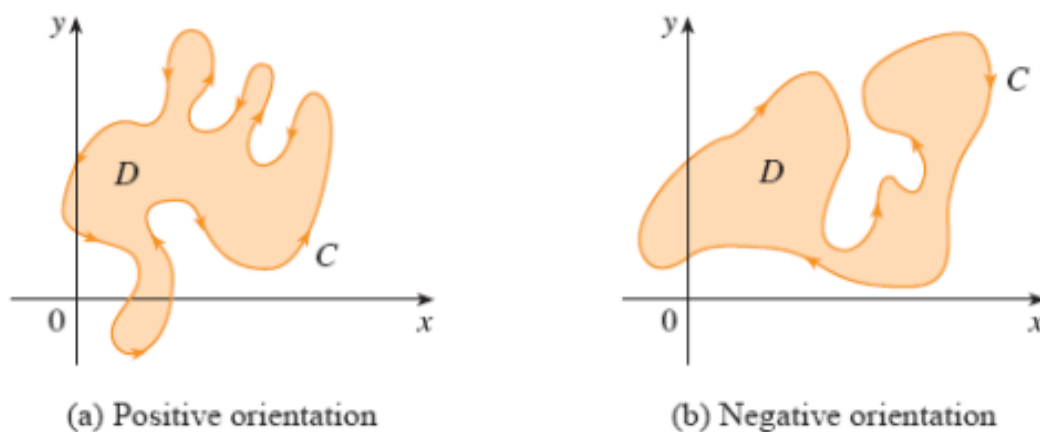


Figure 2.0.1: Oriented curves

Theorem 2.0.3. Green's theorem: Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C Pdx + Qdy = \int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Remark 2.0.4. In the statement of Green's theorem, $dA = dxdy$. Also observe that a line integral becomes a double integral in the statement of Green's theorem. Also, in the statement of Green's theorem, $D \subseteq D'$, where D' is some open region.

Definition 2.0.5. A region D (in $\mathbb{R}^2$) is called a **simple region** if it can be expressed both as type I and type II regions. $\square$

Remark 2.0.6. In Green's theorem, since C is a closed curve, we sometimes also use the symbol $\oint_C$ for the line integral. Moreover, in order to denote that C is the boundary of D , the notation ∂D is used for C .

PROOF: (of Green's theorem): The proof of Green's theorem for general regions is going to be difficult. But we can prove it for simple regions. If we can somehow show that

$$\int_C Pdx = - \int_D \int \frac{\partial P}{\partial y} dA \quad (+)$$

and

$$\int_C Qdy = \int_D \int \frac{\partial Q}{\partial x} dA \quad (++) ,$$

then we are done.

In Stewart's book, in order to prove (+), Stewart considered D to be a type I region, and to prove (++), he considered D as a type II region. This is because a simple region can be expressed both as a type I and type II region. But what could be confusing is that: When he has considered D as a type I region, he has considered D to be of the shape as shown in Figure 2.0.2. That is, D is bounded by the curves C_1, C_2, C_3, C_4 . But even for a region in which there are no vertical curves like C_2 and C_4 , we can workout the proof.

Now if the region D is of the type given in Figure 2.0.2, which is bounded by the curve C , where $C = C_1 \cup C_2 \cup C_3 \cup C_4$, then let us see the proof. So here we have

$$D = \{(x, y) | a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}.$$

Since D is of type I, we have

$$\int_D \int \frac{\partial P}{\partial y} dA = \int_{x=a}^b \int_{y=g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y} dy dx$$

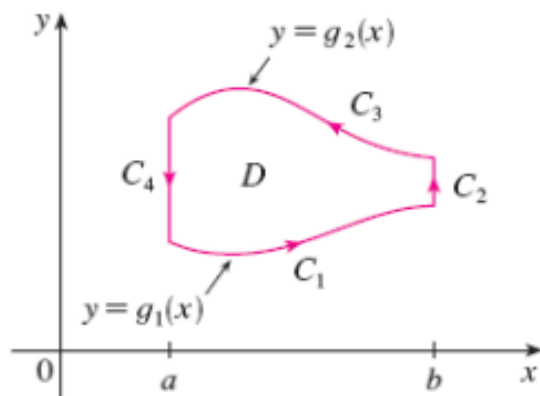


Figure 2.0.2: Proof of Green's theorem

$$= \int_a^b [P(x, g_2(x)) - P(x, g_1(x))] dx. \quad (+ + +)$$

Now

$$\int_C P dx = \sum_{i=1}^4 \int_{C_i} P dx.$$

But along C_2 and C_4 , x is constant. So along these two curves, $dx = 0$. Hence we have

$$\int_C P dx = \int_{C_1} P dx + \int_{C_3} P dx.$$

Now if we consider the curve given by $y = g_2(x)$ from a to b (that is, in the increasing direction of x), then $y = g_2(x)$ will correspond to the curve $-C_3$. That means we can write

$$\begin{aligned} \int_C P dx &= \int_{C_1} P dx - \int_{-C_3} P dx \\ &= \int_a^b [P(x, g_1(x)) - P(x, g_2(x))] dx. \end{aligned}$$

The above expression is nothing but the negative of the expression in $(+ + +)$. So we have proved $(+)$. Similarly, we can prove $(++)$ if we consider D as a type II region. This part of the proof is left to the reader as an exercise. $\square$

Remark 2.0.7. Green's theorem converts a line integral to a double integral or vice versa. Whichever (line/double) integral is easier to compute, we should go for that one.

Example 2.0.8. Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve whose sides are the line segments joining the 3 points $(0,0)$, $(1,0)$, $(0,1)$. Figure 2.0.3 shows the triangular curve C . Let D denote the region bounded by C . Observe that D is a simple region.

By Green's theorem,

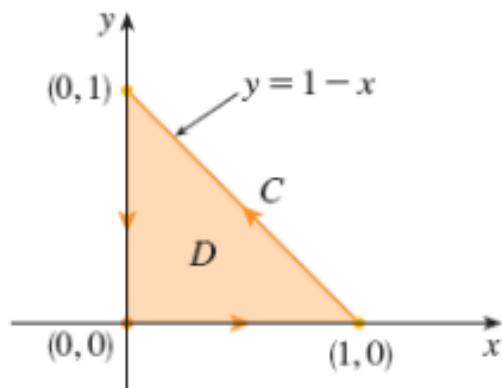


Figure 2.0.3: Figure for Example 2.0.8

$$\begin{aligned} \int_C x^4 dx + xy dy &= \int_D \int \left[\frac{\partial}{\partial x}(xy) - \frac{\partial}{\partial y}(x^4) \right] dA \\ &= \int_D \int y dA = \int_{x=0}^1 \int_{y=0}^{1-x} y dy dx, \end{aligned}$$

where the last equality comes because we are considering D as a type I region. The rest of the calculation is left to the reader as an exercise. $\square$

Example 2.0.9. Evaluate $\int_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy$, where C is the circle given by $x^2 + y^2 = 9$.

By Green's theorem, the above mentioned integral

$$= \int_D \int \left[\frac{\partial}{\partial x}(7x + \sqrt{y^4 + 1}) - \frac{\partial}{\partial y}(3y - e^{\sin x}) \right] dA,$$

where D is the disk enclosed by the circle C . Observe that

$$\frac{\partial}{\partial x}(7x + \sqrt{y^4 + 1}) = 7$$

and

$$\frac{\partial}{\partial y}(3y - e^{\sin x}) = 3.$$

So the given integral becomes

$$= \int_D \int (7-3) dA = \int_D \int 4 dA = 4 \int_D \int 1 dA = 4 \times \text{Area of } D = 4 \cdot \pi(3)^2 = 36\pi.$$

Observe that the integrand is such that in the part $7x + \sqrt{y^4 + 1}$, the portion having x in it can be separated. Similarly, for the other part. Since

$$\frac{\partial}{\partial x}(\sqrt{y^4 + 1}) = 0 = \frac{\partial}{\partial y}(e^{\sin x}),$$

therefore the complicated looking double integral becomes easy to compute. If we don't use Green's theorem, the integration is going to be very difficult. $\square$

2.1 Application of Green's theorem

If we consider a region D bounded by a simple closed curve C , then

$$\int_D \int 1 dA = \text{Area of } D.$$

Now if we want to construct $\int_D \int 1 dA$ in the form $\int_C P dx + Q dy$, then we have to choose P and Q in such a way that

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1.$$

There are many possibilities for P and Q . For instance, we can take

$$P = 0, Q = x$$

or

$$P = -y, Q = 0$$

or

$$P = -\frac{y}{2}, Q = \frac{x}{2}$$

and so on. That's why the area $\int_D \int 1 dA$ becomes $\oint_C x dy$ or $-\oint_C y dx$ or $\frac{1}{2} \oint_C x dy - y dx$ in terms of line integral.

Exercise: Using either of the above 3 formulae for P and Q , find out the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where we use the parametrization

$$x = a \cos t, y = b \sin t, 0 \leq t \leq 2\pi.$$

Remark 2.1.1. From Green's theorem, we obtain a relation between a line integral and a double integral.

Recall that we proved Green's theorem for simple regions (A simple region is a region in the plane which can be expressed as a both type I and type II region.). In this lecture, we are going to discuss about extension of Green's theorem to general regions. The region under consideration need not be simple, but if we are able to express the region as a union of some simple regions, then our job is done.

3 Extension of Green's theorem to general regions

3.1 Green's theorem for regions without holes

Suppose we have a region D (which is not simple and without holes) bounded by a positively oriented, piecewise smooth, simple closed curve C . Now we will divide the region D in such a way that it is a union of 2 simple regions. For doing this, we introduce a vertical line, which divides the region D into 2 parts. See Figure 3.1.1 for an illustration. It can be any curve through which we can divide a region into union of two simple regions. It need not always be a vertical line. Such lines or curves are known as **cross-cuts**.

We can see from Figure 3.1.1 that by introducing the vertical line (drawn in blue color), we have divided the region D into 2 parts D_1 and D_2 . We can also see from the same figure that D_1 and D_2 are simple regions. Then we have

$$\int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{D_1} \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA + \int_{D_2} \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

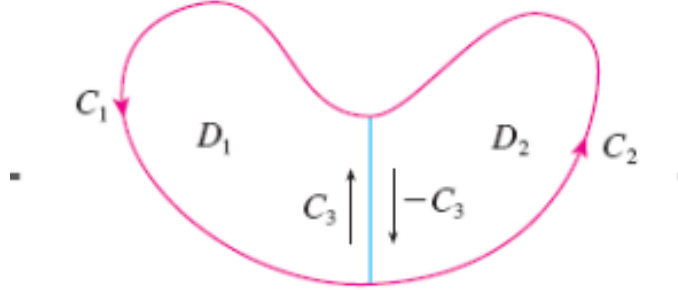


Figure 3.1.1: Green's theorem for a general region

Here $C = C_1 \cup C_2$, where C_1 and C_2 are as shown in Figure 3.1.1. Let us first concentrate on the region D_1 . Then we can see that D_1 is bounded by the union of the curves C_1 and C_3 . Similarly, D_2 is bounded by the union of the curves C_2 and $-C_3$. Then using Green's theorem for simple regions, we get

$$\int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_{C_1 \cup C_3} (Pdx + Qdy) + \oint_{C_2 \cup (-C_3)} (Pdx + Qdy).$$

Using properties of line integrals, we have

$$\begin{aligned} & \int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \\ &= \int_{C_1} (Pdx + Qdy) + \int_{C_3} (Pdx + Qdy) + \int_{C_2} (Pdx + Qdy) + \int_{-C_3} (Pdx + Qdy) \\ &= \int_{C_1} (Pdx + Qdy) + \int_{C_3} (Pdx + Qdy) + \int_{C_2} (Pdx + Qdy) - \int_{C_3} (Pdx + Qdy) \\ &= \int_{C_1 \cup C_2} Pdx + Qdy = \int_C Pdx + Qdy. \end{aligned}$$

That means irrespective of whether or not the region under consideration is simple, if the region is bounded by a positively oriented, piecewise smooth, simple closed curve, we can always get Green's theorem in the same form (as for simple regions).

Example 3.1.1. Evaluate $\oint_C y^2 dx + 3xy dy$, where C is the boundary of the semi-circular region D in the upper half plane between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$. Figure 3.1.2 shows the region D .

Clearly D is not a simple region. Let L denote the vertical cross-cut given

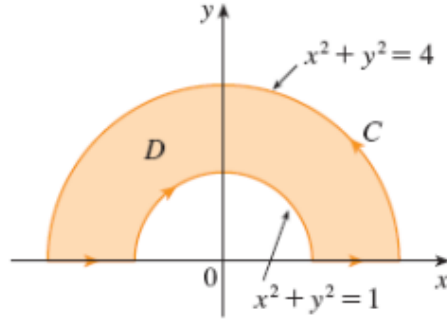


Figure 3.1.2: Figure for Example 3.1.1

by the line segment on the y -axis, which lies inside the region D . Then using the vertical cross-cut L , we can easily express D as a union of 2 simple regions. Now using Green's theorem (for regions which are not simple), we get

$$\oint_C y^2 dx + 3xy dy = \int_D \int \left\{ \frac{\partial}{\partial x}(3xy) - \frac{\partial}{\partial y}(y^2) \right\} dA = \int_D \int y dA.$$

Now if we use polar coordinates, the calculation will become easier. So using the transformations $x = r \cos \theta$, $y = r \sin \theta$, we get that

$$\int_D \int y dA = \int_{r=1}^2 \int_{\theta=0}^{\pi} (r \sin \theta) r dr d\theta.$$

We can see that the variables r and θ in the integrand on the right hand side of the above equation are separable. So we can write

$$\int_{r=1}^2 \int_{\theta=0}^{\pi} (r \sin \theta) r dr d\theta = \int_{r=1}^2 r^2 dr \int_{\theta=0}^{\pi} \sin \theta d\theta.$$

The rest of the calculation is left to the reader as an exercise. □

3.2 Green's theorem for regions with holes

Green's theorem can be extended to regions with holes, that is, regions that are not simply connected. Consider such a region D (as shown in Figure 3.2.1).

Observe that the boundary C of the region D in Figure 3.2.1 consists of two

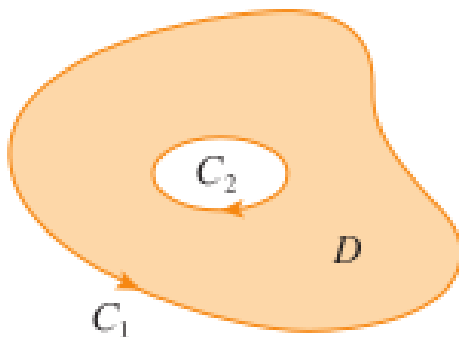


Figure 3.2.1: A region with a hole

simple closed curves C_1 and C_2 . We assume that these boundary curves are oriented in such a way that the region D is always on the left as the curve C is traversed. Thus the positive orientation is anticlockwise for the outer curve C_1 , but clockwise for the inner curve C_2 .

If we divide D into 2 regions D' and D'' by means of the lines shown in Figure 3.2.2 and then apply Green's theorem (the previous version of it) to

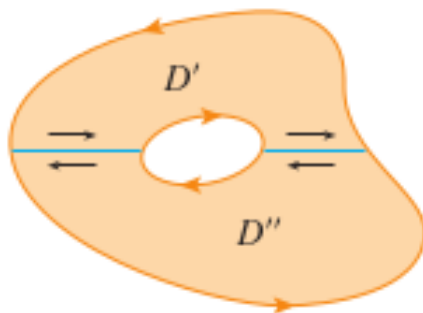


Figure 3.2.2: A region with a hole-having cross cuts

each of D' and D'' , we get

$$\begin{aligned}\int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA &= \int_{D'} \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA + \int_{D''} \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \\ &= \int_{\partial D'} (Pdx + Qdy) + \int_{\partial D''} (Pdx + Qdy),\end{aligned}$$

where $\partial D'$ and $\partial D''$ denote the boundaries of D' and D'' respectively. Since the line integrals along the common boundary lines are in opposite directions, they cancel and we get

$$\int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{C_1} (Pdx + Qdy) + \int_{C_2} (Pdx + Qdy) = \int_C Pdx + Qdy,$$

which is Green's theorem for the region D with a hole.

Remark 3.2.1. So we can apply Green's theorem to regions with holes as well. But we have to be careful about the orientation of the inner closed curve.

Example 3.2.2. If $\vec{F} = -\frac{y}{x^2+y^2}\hat{i} + \frac{x}{x^2+y^2}\hat{j}$, then show that $\oint_C \vec{F} \cdot d\vec{r} = 2\pi$, for every positively oriented simple closed path that encloses the origin.

Observe that we are not given any region D here. We know that if $\vec{F} = P\hat{i} + Q\hat{j}$, then $\vec{F} \cdot d\vec{r} = Pdx + Qdy$. That means we have to evaluate $\int_C Pdx + Qdy$, where $P = -\frac{y}{x^2+y^2}$ and $Q = \frac{x}{x^2+y^2}$.

Now we are going to introduce a very small circle with radius ϵ in such a way that this small circle is enclosed by the original curve C . Let us denote the small inner circle by C' . Please see Figure 3.2.3 for an illustration.

Let us take anticlockwise orientation for both C and C' . Consider the closed curve $-C'$, and the region D enclosed by C and $-C'$ (both C and $-C'$ are positively oriented with respect to the region D that they enclose). Then by Green's theorem for regions with holes, we have

$$\int_{C \cup (-C')} Pdx + Qdy = \int_D \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

In our case, $P = -\frac{y}{x^2+y^2}$ and $Q = \frac{x}{x^2+y^2}$. So one can easily check that

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

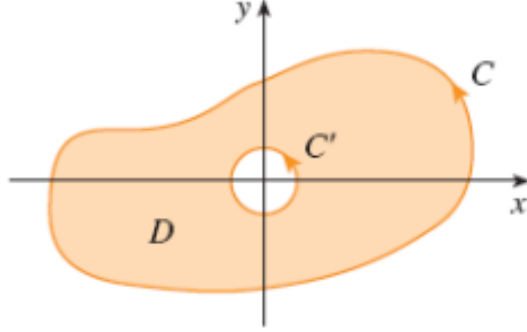


Figure 3.2.3: Figure for Example 3.2.2

That implies

$$\int_{C \cup (-C')} Pdx + Qdy = 0.$$

Or in other words,

$$\begin{aligned} \int_C (Pdx + Qdy) - \int_{C'} (Pdx + Qdy) &= 0 \\ \Rightarrow \int_C Pdx + Qdy &= \int_{C'} Pdx + Qdy. \end{aligned}$$

Now we will use polar coordinates for computing the integral on the right hand side of the above equation. That means, applying the transformations $x = \epsilon \cos \theta$, $y = \epsilon \sin \theta$, we get

$$\begin{aligned} \int_C Pdx + Qdy &= \int_{C'} Pdx + Qdy = \int_{\theta=0}^{2\pi} \left[\frac{\epsilon^2 \sin^2 \theta}{\epsilon^2} + \frac{\epsilon^2 \cos^2 \theta}{\epsilon^2} \right] d\theta \\ &= \int_{\theta=0}^{2\pi} d\theta = 2\pi. \end{aligned}$$

□

4 Curl and divergence

4.1 Curl of a vector field

Let $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$ be a vector valued function, where P, Q, R have first order partial derivatives with respect to x, y, z in some domain $D \subseteq \mathbb{R}^3$.

Then the **curl** of $\vec{F}$ is denoted by $\text{curl}\vec{F}$ and is defined as

$$\text{curl}\vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \hat{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k}.$$

Previously we had defined the gradient vector as

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}.$$

This can also be written as

$$\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) f.$$

Let us define $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$ to be the **gradient operator**. Then it is clear that

$$\text{curl}\vec{F} = \nabla \times \vec{F}$$

because

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

Curl has a lot of applications in fluid dynamics. Curl can be associated with the rotation of a fluid. Let the velocity of a fluid in a certain domain be $\vec{F}$. Then the curl $\nabla \times \vec{F}$ is always going to be along the axis about which the fluid is going to rotate. Please see Figure 4.1.1 for this.

Definition 4.1.1. Let $\vec{F}(x, y, z)$ denote the velocity of a fluid in a certain domain at (x, y, z) . We already know that $\text{curl}\vec{F}$ designates the rotation of the fluid. Then if $\nabla \times \vec{F} = \vec{0}$, we say that the fluid is **irrotational** and if $\nabla \times \vec{F} \neq \vec{0}$, we say that the fluid is **rotational**. $\square$

Theorem 4.1.2. If f is a function of 3 variables that has continuous second order partial derivatives, then

$$\text{curl}(\nabla f) = \vec{0}.$$

PROOF: We have

$$\text{curl}(\nabla f) = \nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}$$

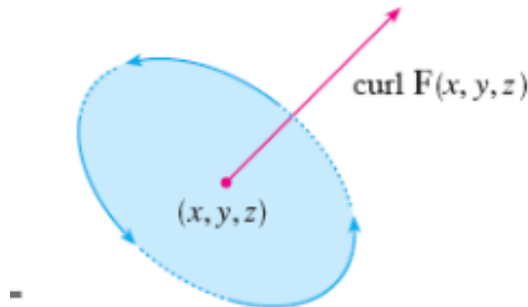


Figure 4.1.1: A rotating fluid

$$\begin{aligned}
 &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \hat{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \hat{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \hat{k} \\
 &= 0\hat{i} + 0\hat{j} + 0\hat{k} = \vec{0}. \text{ (By Clairaut's theorem)}
 \end{aligned}$$

□

Remark 4.1.3. Since a conservative vector field is one for which $\vec{F} = \nabla f$, therefore we have

$$\text{If } \vec{F} \text{ is conservative, then } \text{curl} \vec{F} = \vec{0}.$$

This gives us a way of verifying that a vector field is not conservative.

4.2 Divergence of a vector field

Let $\vec{F} = \langle P, Q, R \rangle$ be a given vector field. Assume that P, Q, R possess continuous first order partial derivatives. Then the **divergence** of $\vec{F}$ is denoted by $\nabla \cdot \vec{F}$ and is defined as

$$\nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Definition 4.2.1. If $\nabla \cdot \vec{F} = 0$ for any vector field $\vec{F}$, then we say that the vector field is **solenoidal**. □

Definition 4.2.2. Let $\vec{F}(x, y, z)$ denote the velocity of a fluid in a certain domain at (x, y, z) . If $\nabla \cdot \vec{F} = 0$, then we say that the fluid is **incompressible**.
□

Theorem 4.2.3. If $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$ is a vector field on $\mathbb{R}^3$ and P, Q, R have continuous second order partial derivatives. Then

$$\nabla \cdot (\nabla \times \vec{F}) = 0.$$

PROOF: Since

$$\nabla \times \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \hat{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k},$$

therefore

$$\nabla \cdot (\nabla \times \vec{F}) = \frac{\partial}{\partial x} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = 0,$$

by Clairaut's theorem (because by hypothesis of the theorem, we have that P, Q, R have continuous second order partial derivatives). □

Remark 4.2.4. We sometimes also use the notation $\text{div} \vec{F}$ to denote divergence of a vector field $\vec{F}$.

Remark 4.2.5. Curl is a vector function, but divergence is a scalar function. Whenever we write that $\text{curl} \vec{F}$ is zero, we use the vector notation $\vec{0}$. And scalar notation 0 for divergence.

—————-END—————